STUDY BUDDY STEM AI: AN AI-POWERED PERSONALIZED, ADAPTIVE AND MULTIMODAL STEM LEARNING ECOSYSTEM FOR SRI LANKAN STUDENTS

COMMON INTEGRATED SOLUTION REPORT

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Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Students preparing for the Sri Lankan G.C.E. Ordinary Level examination learn STEM subjects mainly through textbooks, teacher-led lessons and fixed assessments. This leaves little room for personal explanation, practice at the right level, or help outside the classroom. General AI chatbots are available, but they are not tied to the national syllabus and can state facts that are wrong.

This report presents Study Buddy STEM AI, a single web application that joins four components built by four team members. Personalized Storytelling teaches a topic through a short story matched to the student. Conversational AI answers typed or spoken questions. Adaptive Assessment runs a quiz that responds to the student's answers, behaviour and facial expression. Knowledge Maps checks a student's handwritten notes and turns them into revision material. All four draw on the same official Grade 10 and 11 Science textbooks through retrieval-augmented generation, and three of them share one model runtime.

Each component was built and evaluated separately and then joined in one application. The storytelling component produced correctly structured lessons in all 12 benchmark cases, with a quality score of 91.7%. Qwen2.5 7B scored highest of three models tested for the tutor, at 88.6. The selected facial-expression model processed a frame in 27.7 milliseconds on a CPU. Gemma 4 scored highest of three vision-language models tested for reading notes, at 88.6. Testing of the complete solution with students is still to be done.

The work shows that one curriculum-grounded application can support a student through learning, asking, testing and reviewing a topic, using mostly openly available models.

Keywords: STEM education, retrieval-augmented generation, personalised learning, adaptive assessment, G.C.E. Ordinary Level

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
CPU	Central processing unit
FAISS	Facebook AI Similarity Search, a vector search library
G.C.E. O/L	General Certificate of Education, Ordinary Level
JSON	JavaScript Object Notation
LLM	Large language model
ML	Machine learning
NIE	National Institute of Education
OCR	Optical character recognition
PDPA	Personal Data Protection Act
RAG	Retrieval-augmented generation
RRF	Reciprocal rank fusion
SDG	Sustainable Development Goal
STEM	Science, technology, engineering and mathematics
TTS	Text to speech
YOLO	You Only Look Once, an object detection model

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

STEM education builds scientific reasoning, problem solving and technical skill. In Sri Lanka, students meet these subjects in a demanding form at the G.C.E. Ordinary Level (O/L), which covers Grades 10 and 11 and ends in a national examination.

Teaching at this level relies mostly on the textbook, on explanations led by the teacher, and on the same assessments for every student. Within this setting our project identified six recurring difficulties: low engagement, trouble understanding abstract concepts, little adaptive support, stress around assessment, poor retention, and limited access to personal tutoring and revision help.

Research shows that artificial intelligence (AI) can help with each of these. Reviews of AI in STEM education report its use for personalised content, tutoring, feedback and the analysis of learner behaviour [1]-[4]. Two problems remain. The first is accuracy. A general language model can give a fluent answer that is wrong or outside the syllabus. Retrieval-augmented generation (RAG) reduces this risk by giving the model passages retrieved from a trusted source to write from [5]. The second is fragmentation. Most systems offer one capability, such as tutoring or assessment, so a student has to move between unrelated tools.

Study Buddy STEM AI responds to both problems. It joins four AI components in one application, and it grounds them in the official textbooks. The solution is designed for STEM learning in general. Because this research had to be completed within one year, we limited its scope to one subject, Science, for Grades 10 and 11.

1.2 Stakeholders and User Needs

Table 1.1 lists the stakeholders of the solution and what each of them needs from it.

Table 1.1: Stakeholders and their needs

Stakeholder	Role	Main needs
Students in Grades 10 and 11	Primary users	Explanations they can follow, help at any time, practice at the right level, and a way to check their own notes

Stakeholder	Role	Main needs
Teachers and tuition teachers	Secondary users	Material that follows the syllabus, a view of how students are doing, and a tool that supports their teaching
Parents	Secondary stakeholders	Affordable support at home, and assurance that the content is correct and safe
Schools and tuition institutes	Adopters	Low running cost, simple installation, and protection of student data
Curriculum bodies	Content owners	Faithful use of the official textbooks

Our direct evidence of user needs comes from an anonymous survey of 28 students in Grades 10 and 11, carried out for the storytelling component. Nineteen of the 28 students said that at least some parts of school science are hard for them. Their preferred story styles were spread across all five options offered, which supports the need for personalisation and shows that one style does not suit every student.

-PENDING [Describe any engagement with teachers, parents or schools, such as interviews, demonstrations or feedback sessions. If there was none, state that engagement so far has been with students only.]

1.3 Alignment with the Sustainable Development Goals

The project aligns with Sustainable Development Goal 4, which is to ensure inclusive and equitable quality education and promote lifelong learning opportunities for all [8]. Two of its targets are most relevant.

- Target 4.1 concerns the completion of quality primary and secondary education that leads to effective learning outcomes. The solution is intended to help secondary students understand and retain Science content.
- Target 4.5 concerns equal access to education. The solution is intended to give students who cannot afford private tuition a form of personal support.

These are intended contributions. We have not yet measured any effect on learning outcomes or on access, so we make no claim of impact. Section 7.5 discusses what would be needed to show one.

1.4 Purpose and Scope of the Common Report

This report presents the common problem and the integrated solution. It explains how the four components relate, how they were joined, how the complete solution was evaluated and what value it offers.

The report does not repeat the detailed research behind each component. Each team member's individual thesis gives the literature, design, implementation and experiments for one component. Where this report gives a component-level result, it does so in summary and points to the individual thesis for the detail.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

The common problem is as follows. Students preparing for the G.C.E. O/L examination do not have access to learning support that is personal to them, accurate to the national syllabus, and available throughout the stages of studying a topic.

A student studies a topic in four stages: understanding it, asking questions about it, testing themselves on it, and reviewing their own notes. At each stage the support available today is the same for everyone, is limited by the teacher's time, or is not tied to the syllabus.

2.2 Root Causes and Overall Research Gap

The problem has five main causes.

- A textbook explains each topic in one way, although students differ in their interests and in how much they struggle.
- A teacher has limited time, so few students can get a personal explanation or ask every question they have.
- Assessments use a fixed set of questions at a fixed level, and take no account of how the student is coping.
- Students write their own notes, but nobody checks those notes against the textbook.
- General AI chatbots are not tied to the Sri Lankan syllabus, and can give answers that are wrong or out of scope.

Existing AI systems address these causes one at a time. Research covers personalised content, intelligent tutoring, adaptive assessment and emotion-aware learning as separate systems [1]-[4], [7]. A student who wants all of them must use several tools that share no content and no context.

The overall gap is therefore one of integration and grounding. To our knowledge, no single system supports a Sri Lankan O/L student through all four stages of studying a topic while drawing its content from the official textbooks. Closing this gap needs

four components, one for each stage, built on a shared base of curriculum knowledge.

2.3 Subproblem and Component Mapping

We divided the common problem into four subproblems, one for each stage of studying a topic. Table 2.1 maps each subproblem to its component.

Table 2.1: Subproblems and components

Stage	Subproblem	Component	Owner
Learn	Abstract concepts are hard to connect with a student's own life, and one explanation does not suit everyone	RAG-Based Personalized STEM Storytelling	Elwitigala S.A.I.
Ask	Students cannot get an immediate, syllabus-based answer to a question outside class, in a language and form that suits them	Multilingual Conversational AI	Manodya K.M.B.
Test	Fixed assessments ignore how the student is coping, and cause stress	Emotion-Aware Adaptive Assessment	Basnayaka W.B.M.D.K.
Review	A student's handwritten notes are not checked or turned into revision material	Handwritten Content Analysis and Synthesis	Dilshan W.A.

The four subproblems do not overlap, because each concerns a different stage and a different kind of input: a survey and a topic, a question, a set of quiz answers, and a photograph of notes. They depend on each other in two ways. All four rely on the same curriculum knowledge, so that a student meets consistent content at every stage. Together they cover the whole study cycle, so removing any one of them leaves a stage without support.

2.4 Overall Aim and Group Objectives

The overall aim is to design, build and evaluate one AI-supported learning application that helps Sri Lankan O/L students to learn, ask about, test themselves on and review Science topics, with content taken from the official textbooks.

Table 2.2 lists the group objectives, the components each one involves, and the evidence used to judge it.

Table 2.2: Group objectives

Objective	Components	Evidence
G1. Build a shared base of curriculum knowledge from the official Grade 10 and 11 Science textbooks, and make it searchable	All	The indexes built, and retrieval results in the components
G2. Build four working components, one for each stage of studying a topic	All	The results of each component
G3. Join the four components in one web application with shared services	All	Integration and end-to-end tests
G4. Use openly available models through a common runtime where possible, to limit running cost	All	The models selected and the results of the model comparisons
G5. Evaluate each component and the complete solution	All	Component results, integrated tests and user validation

2.5 Scope, Assumptions and Constraints

Scope. The solution covers Science for Grades 10 and 11, using the official English-medium textbooks. The tutor supports English and Sinhala. The solution is a study aid that supports the teacher and does not replace the teacher.

Exclusions. The solution does not cover other STEM subjects, does not mark examination answers, and has not been deployed in a school.

Assumptions. We assume that the official textbooks are correct, that a student can read English at the level of the textbook, and that a student answers surveys and quizzes honestly.

Constraints. The project had to be completed within one year by four students. The solution runs on one computer that must be able to run the language models. Adaptive Assessment needs a webcam.

Dependencies. Three parts of the solution need an internet connection: the quiz database, the vision-language model used by Knowledge Maps, and the optional speech services of the tutor. The storytelling component and the core of the tutor run on the local computer.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

Study Buddy STEM AI is one web application. A student opens it in a browser and reaches all four components from the same place. The components are built on the same textbooks, so the content a student meets in a story, in the tutor's answer, in a quiz question and in a note check comes from one source.

The value of the solution is this continuity. A student can learn a topic through a story, ask follow-up questions, test themselves, and then check their own notes, without leaving the application and without the content changing between tools. No single component offers this by itself.

3.2 High-Level Architecture

Figure 3.1 shows the architecture of the complete solution.

The architecture has four parts. Students and teachers use the solution through a web browser. The browser sends requests to a FastAPI web application, which holds the four components. The components use three shared services on the same computer: the curriculum data, the Ollama runtime that serves the language models, and a separate machine learning service that generates quiz questions. A dashed line marks the deployment boundary of that computer.

Three kinds of external service lie outside the boundary and need an internet connection. The quiz stores its data in MongoDB Atlas. Knowledge Maps uses the Gemma 4 model through a cloud tag in Ollama. The remaining services are optional: speech services for the tutor, a fallback vision model, and video transcripts.

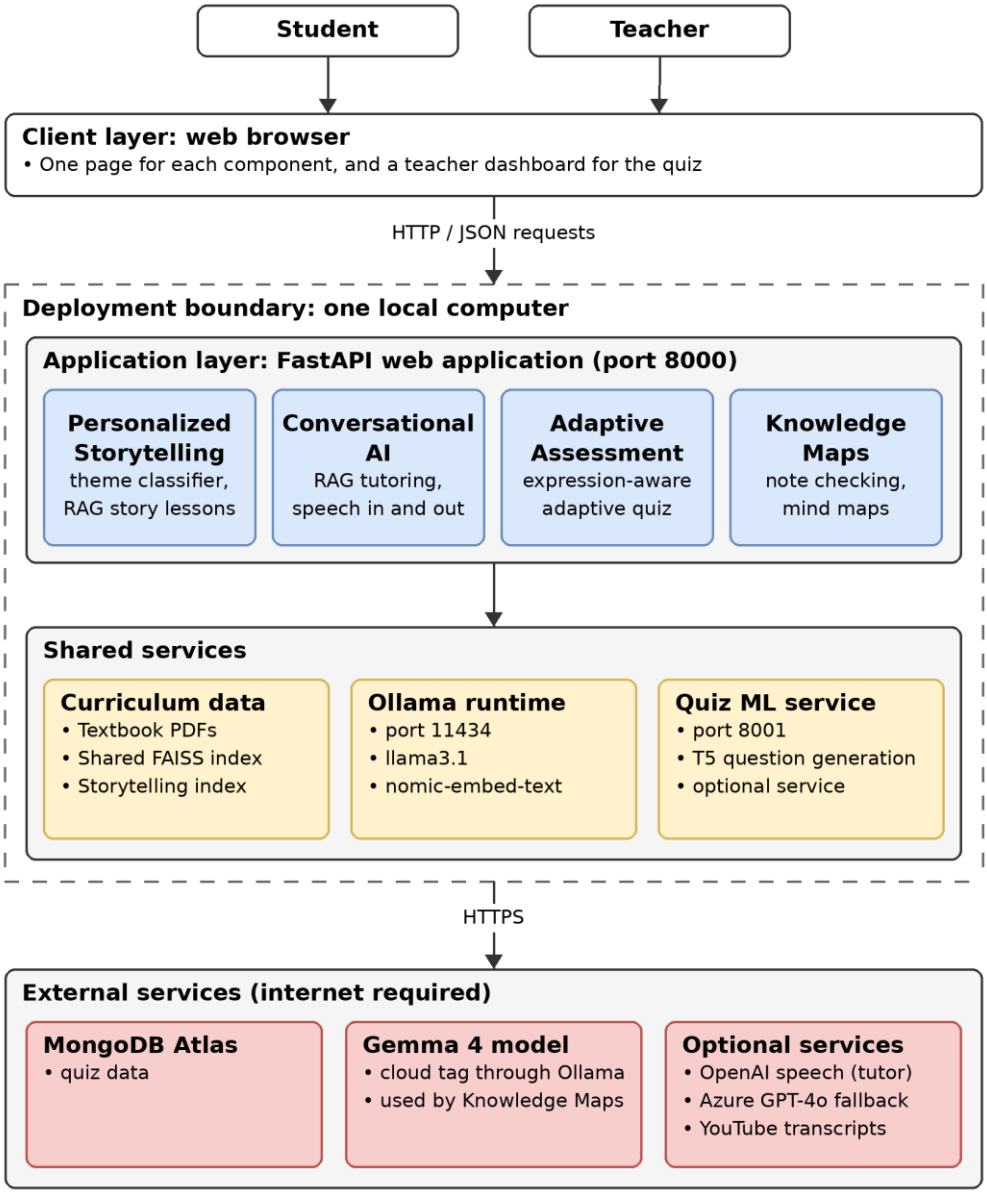


Figure 3.1: High-level architecture of the complete solution

3.3 Component 1 Summary

Table 3.1 summarises the first component.

Table 3.1: Summary of RAG-Based Personalized STEM Storytelling

Item	Description
Owner	**Elwitigala S.A.I. – IT22087638**
Subproblem	Abstract concepts are hard to connect with a student's own life, and one explanation does not suit everyone

Item	Description
Input	Six survey answers from the student, and a topic chosen from the textbook
Core method	A Random Forest classifier predicts a story theme from the student's interest and hoped-for job. Passages on the topic are retrieved from the textbook. Llama 3.1, run through Ollama, writes the lesson from those passages.
Output	A lesson in two parts: a plain explanation, then a three-paragraph story matched to the student, with definitions, equations, exam points and the source passages
Contribution to the solution	Supports the Learn stage. It gives each student a first explanation that fits their interests and stays within the textbook.

3.4 Component 2 Summary

Table 3.2 summarises the second component.

Table 3.2: Summary of Multilingual Conversational AI

Item	Description
Owner	Manodya K.M.B. - IT22136756
Subproblem	Students cannot get an immediate, syllabus-based answer to a question outside class
Input	A question typed or spoken by the student, in English or Sinhala
Core method	Speech is transcribed with Whisper. Passages are retrieved from the shared textbook index. A language model writes the answer from those passages and keeps the history of the conversation. The answer can be read aloud
Output	An answer based on the textbook, as text and as speech
Contribution to the solution	Supports the Ask stage. It lets a student follow up on anything they did not understand, at any time.

3.5 Component 3 Summary

Table 3.3 summarises the third component.

Table 3.3: Summary of Emotion-Aware Adaptive Assessment

Item	Description
Owner	Basnayaka W.B.M.D.K. - IT22272768
Subproblem	Fixed assessments ignore how the student is coping, and cause stress
Input	The student's answers, the time taken, changes of answer, the quiz level, and the student's facial expression from the webcam
Core method	A YOLOv8s model detects the facial expression. A Random Forest model combines the signals into a learning state, which sets the feedback and the next level of difficulty. Questions can also be generated from a PDF by a pre-trained T5 model in a separate service.
Output	Adaptive feedback, the next question at a suitable level, and a dashboard for the teacher
Contribution to the solution	Supports the Test stage. It lets a student check their understanding at a level that responds to how they are coping.

3.6 Component 4 Summary

Table 3.4 summarises the fourth component.

Table 3.4: Summary of Handwritten Content Analysis and Synthesis

Item	Description
Owner	Dilshan W.A. - IT22292568
Subproblem	A student's handwritten notes are not checked or turned into revision material
Input	A photograph of the student's handwritten notes, or the transcript of a video

Item	Description
Core method	The image is cleaned and read by a vision-language model, Gemma 4, through Ollama. The content is checked against the textbook using a hybrid search that combines vector search, keyword search and a reranker. Content is accepted only when enough supporting evidence is found.
Output	Verified notes, mind maps, flashcards and structured notes
Contribution to the solution	Supports the Review stage. It turns a student's own notes into checked revision material.

3.7 End-to-End User and Data Flow

Figure 3.2 shows how a student moves through the complete solution.

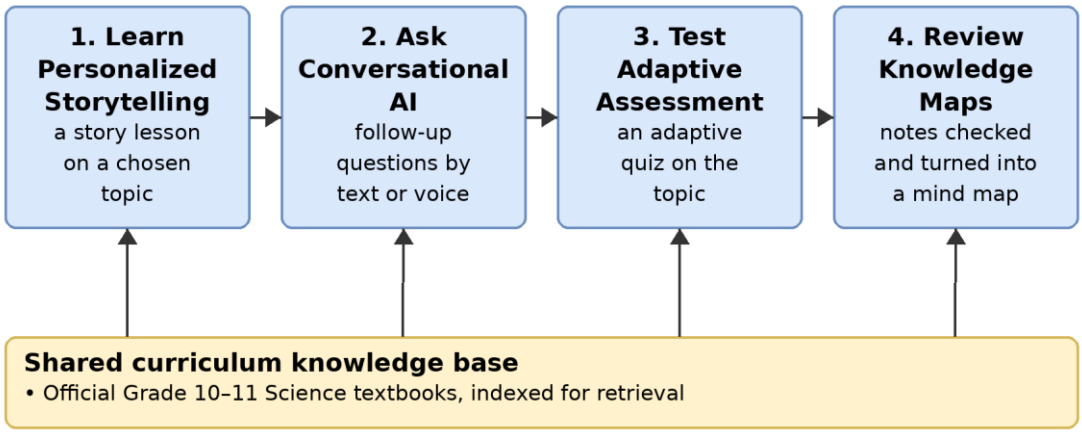


Figure 3.2: End-to-end learning flow

A typical session has four steps. The student first completes a short survey and selects a topic, and Personalized Storytelling returns a lesson. The student then asks the Conversational AI about any part that was unclear. Next, the student takes an adaptive quiz on the topic. Finally, the student photographs their own notes on the topic and Knowledge Maps checks them and builds a mind map.

The data that connects the steps is the curriculum knowledge base. Every component retrieves from the same textbooks, so the explanation, the answers, the questions and the note check agree with each other.

At present the student moves from one step to the next through the navigation of the application.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

We integrated the components by building them as separate modules of one web application. Each component has its own set of requests and its own page, and the application registers all four when it starts. This approach let each member develop and test a component alone, and then join it without rewriting it.

The integration followed three steps. We first agreed the shared resources: the textbook files, the format of the shared index and the model runtime. Each member then built a component against those resources. Finally we registered the components in the common application and tested them together.

Each member owns the interface of their own component. The shared resources are owned by the team. Table 4.1 lists the main integration risks and how we managed them.

Table 4.1: Integration risks and how they were managed

Risk	How it was managed
Components need different versions of the same library	One project environment with one agreed set of library versions
Keys and settings are exposed in the repository	A settings file that is excluded from the repository and shared separately
Large index files cannot be stored in the repository	A shared pack that holds the settings file and the indexes for every member
A change in one component breaks another	Separate modules with separate requests, so that a component can change without affecting the others

4.2 Interfaces, Data Contracts and Shared Models

All components communicate with the browser in the same way, through requests and responses in JSON over HTTP. Each component has its own address prefix within the application. Table 4.2 lists the interfaces.

Table 4.2: Component interfaces

Component	Address prefix	Main requests
RAG-Based Personalized STEM Storytelling	/narrative-learning	Status, survey, classify, chapters, generate, theme feedback
Multilingual Conversational AI	/voice-tutor	Ask a question, transcribe speech, read answer aloud
Emotion-Aware Adaptive Assessment	/adaptive-quiz	Sign in, get quiz questions, submit answer, detect expression, predict learning state, generate questions from a PDF, teacher dashboard
Handwritten Content Analysis and Synthesis	/knowledge-maps	Check notes, read handwritten notes, build flashcards, build mind map, suggest videos

The components share data and models instead of calling each other. Table 4.3 lists the shared resources.

Table 4.3: Shared data and models

Resource	Description	Used by
Textbook files	The official Grade 10 and 11 Science pupil books and teacher guides, as PDF files	All components
Shared textbook index	Passages of about 200 to 400 words, each tagged with grade and topic, embedded with all-MiniLM-L6-v2 and stored in FAISS	Conversational AI and Knowledge Maps, and Storytelling as a backup
Ollama runtime	One installation that serves the language models on the local computer	Storytelling, Conversational AI and Knowledge Maps
Settings file	Keys, model names and switches for all components	All components

Each component validates its own input and handles its own errors, as described in Section 4.4. The requests of the storytelling component are specified in Appendix A, with space for the other three.

4.3 Shared Non-Functional Requirements

Table 4.4 lists the non-functional requirements that apply to the whole solution and the evidence we have for each.

Table 4.4: Shared non-functional requirements

Area	Requirement	Evidence so far
Performance	Interactive requests respond quickly, and model output arrives in an acceptable time	Storytelling classifies a theme in about 30 milliseconds and generates a lesson in about 44 seconds. The expression model processes a frame in 27.7 milliseconds.
Usability	A student can use every component without training	■ Pending
Security	Keys and indexes are kept out of the repository	Confirmed by automated tests in the storytelling component
Privacy	No more personal data is collected than is needed	The storytelling survey collects no names.
Reliability	A failure in a model does not leave the student without a result	Storytelling returned a valid lesson in 12 of 12 cases and has a fallback
Maintainability	A component can be changed without affecting the others	Each component is a separate module
Scalability	The solution can serve more than one student	Not yet tested. The solution currently runs on one computer.
Accessibility	Students can use speech as well as text	The tutor accepts spoken questions and can read answers aloud

4.4 Integrated Workflow and Exception Handling

In normal operation the model runtime is started first and the web application second. A student then opens the application and follows the flow described in Section 3.7.

Table 4.5 lists the main failure paths and what the solution does in each.

Table 4.5: Failure paths and fallbacks

Failure	Component	Behaviour
The trained classifier cannot be loaded	Storytelling	The theme is taken from the student's own story preference answers
The language model returns an unusable lesson	Storytelling	The request is tried again, and then a lesson built from the textbook text is shown
Too few passages are found in the component's own index	Storytelling	Passages are added from the shared textbook index
No key is set for the cloud speech service	Conversational AI	The local model writes the answer, and the browser reads it aloud
Not enough evidence is found for a statement in the notes	Knowledge Maps	The statement is not accepted
The main vision model is not available	Knowledge Maps	An optional fallback model can be used
The question generation service is not running	Adaptive Assessment	Question generation from a PDF is not available, and the quiz itself continues

4.5 Team Contribution and Dependency Matrix

Table 4.6 shows the work of each member. Each member built one component, and the shared work was divided between the team.

Table 4.6: Team contribution matrix

Member	Component	Research and development	Shared and integration work
Elwitigala S.A.I.	RAG-Based Personalized STEM Storytelling	Survey, theme classifier, textbook retrieval, lesson generation, benchmark and tests	Joined the storytelling component to the common application; linked it to the shared textbook index as a backup source; compiled the common report
Manodya K.M.B.	Multilingual Conversational AI	Speech input, retrieval, answer generation, conversation memory, model comparison	Joined the tutor to the common application; prepared the shared textbook index used by the tutor, Knowledge Maps and storytelling
Basnayaka W.B.M.D.K.	Emotion-Aware Adaptive Assessment	Expression detection, learning-state model, adaptive quiz, question generation service, teacher dashboard	Joined the quiz to the common application; set up the quiz database and sign-in; ran the separate question generation service

Member	Component	Research and development	Shared and integration work
Dilshan W.A.	Handwritten Content Analysis and Synthesis	Note reading, hybrid retrieval, verification, mind maps and flashcards, model comparison	Joined Knowledge Maps to the common application; set up the vision-language model through the shared model runtime and the optional fallback model

The components depend on the shared resources as follows. Storytelling, Conversational AI and Knowledge Maps depend on the Ollama runtime. Conversational AI and Knowledge Maps depend on the shared textbook index, and Storytelling uses it as a backup. Adaptive Assessment depends on its own database and question generation service. Appendix B records the contribution of each member as agreed by the team.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

The components are built on one technology stack. Table 5.1 lists the common parts.

Table 5.1: Common technology stack

Layer	Technology	Version
Language	Python	3.13
Web application	FastAPI with Uvicorn	0.141.1 and 0.52.4
Text extraction	PyMuPDF	1.24 or later
Embeddings for the shared index	sentence-transformers, model all-MiniLM-L6-v2	6.0.0
Vector search	FAISS	1.15.0
Model runtime	Ollama	3.1
Quiz database	MongoDB Atlas with Beanie	2.2.0
Quiz interface	React with Vite	8.0.10

We chose these for three reasons. Python has mature libraries for every method the components use. FastAPI lets each component be written as a separate module of one application. Ollama gives all components the same way to run a language model on the local computer, with no charge for each request.

5.2 Development and Integration Process

-pending

5.3 Deployment and Operational Configuration

The solution is deployed on one computer. It has not been deployed to a public server or to a school.

To run the solution, a member installs the project environment and Ollama, downloads the models, and places the shared pack in the project folder. The pack holds the settings file and the indexes, which are not in the repository. The model

runtime is started first, then the web application, and then the question generation service if it is needed.

The settings file controls which models and services each component uses. The storytelling component can report whether it is ready and whether its trained classifier is loaded.

The indexes can be rebuilt from the textbook files at any time, so they need no separate backup.

5.4 Evidence of Team Execution

Table 5.2 summarises the evidence of how the team carried out the project. The records themselves are kept by the team and are available on request.

Table 5.2: Evidence of team execution

Source	What it shows	Summary
Task board	Planning and progress of tasks	
Meeting records	Decisions and supervision	
Git history	Development activity of each member	
Research diaries	Individual research activity	

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

We evaluated the solution at two levels. Each component was evaluated against the question that matters most for its stage. The complete solution is evaluated for whether the components work together and whether students find it useful.

Table 6.1 sets out the evaluation of each component.

Table 6.1: Evaluation of each component

Component	Question	Method	Measures
RAG-Based Personalized STEM Storytelling	Are the lessons complete, well structured and close to the textbook?	A benchmark of 12 test cases, also run with a baseline model	Quality score, valid output rate, similarity to the textbook, story length, time
Multilingual Conversational AI	Which language model suits tutoring best?	A comparison of three models on five criteria	Score out of 100 for each criterion
Emotion-Aware Adaptive Assessment	Can facial expression be detected fast enough for live use?	A comparison of two detection models on a CPU	Time to process one frame
Handwritten Content Analysis and Synthesis	Which vision-language model suits reading notes best?	A comparison of three models on eight criteria	Score out of 100 for each criterion

For the complete solution, the success criteria are that all four components run in one application, that each one reaches the shared resources it needs, and that students can complete the full learning flow. Detailed experiments for each component are in the individual theses.

6.2 Integration and End-to-End Testing

Table 6.2 shows the tests relevant to the complete solution and their status.

Table 6.2: Integration and end-to-end tests

Test	What it checks	Status
Component inside the application	Storytelling, running in the complete application, reports that it is ready, that its trained classifier is loaded and that four textbooks are available	Passed
Non-functional tests	Fifteen automated checks of response time, configuration, privacy and security in the storytelling component	15 of 15 passed
Shared model runtime	Storytelling generated all 12 benchmark lessons through the shared Ollama runtime	Passed
End-to-end flow	One student completes the four steps in turn on the same topic	[To be completed]
Other components inside the application	Each of the other three components responds correctly in the complete application	[To be completed]
Usability and user acceptance	Students use the complete solution and give feedback	[To be completed]

6.3 Overall Results

This section gives the main result of each component. The individual theses give the full experiments.

Personalized Storytelling. Table 6.3 shows the results of the lesson benchmark. Llama 3.1 returned every lesson in the required format and was better than the baseline model on every measure. The theme classifier predicted the expected theme in 8 of the 12 benchmark cases, and reached [76.1%: confirm after retraining] accuracy in cross-validation.

Table 6.3: Results of the storytelling benchmark

Measure	Llama 3.1	Llama 2 7B (baseline)
Generation quality score	91.7%	63.9%
Output in the required format	12 of 12	8 of 12
Similarity to the textbook	78.0%	62.4%
Average story length	230.9 words	158.4 words
Average time for one lesson	43.8 seconds	63.9 seconds

Conversational AI. Table 6.4 shows the comparison of three language models. Mistral 7B gave the best answers but was the slowest. Llama 3.2 3B was the fastest but gave weaker answers. Qwen2.5 7B had the highest overall score because it was balanced across all five criteria, and it was selected.

Table 6.4: Comparison of language models for the tutor (scores out of 100)

Criterion	Mistral 7B	Qwen2.5 7B	Llama 3.2 3B
Response quality	92	89	78
Context relevance	91	90	76
Explanation quality	93	90	74
Response speed	65	88	95
Resource efficiency	72	86	94
Overall	82.6	88.6	83.4

Adaptive Assessment. Table 6.5 shows the comparison of two expression detection models. YOLOv8s was more than twice as fast as YOLOv8m on a CPU, which makes it suitable for use during a live quiz, and it was selected.

Table 6.5: Comparison of expression detection models on a CPU

Model	Time for one frame	Frames in one second
YOLOv8s (selected)	27.7 milliseconds	About 36
YOLOv8m	59.6 milliseconds	About 17

Knowledge Maps. Table 6.6 shows the comparison of three vision-language models. Gemma 4 had the highest average score and was selected. Its strongest scores were for ease of deployment (95), output structure (92) and consistency (90).

Table 6.6: Comparison of vision-language models for reading notes (average score out of 100)

Model	Average score
Gemma 4 (selected)	88.6
Qwen 2.5	85.5
Llama 3.2	82.4

Taken together, the results show that each component has a model that suits its task, and that three of the four selected models are openly available models that run through the common runtime.

6.4 Stakeholder and User Validation

[This section is completed after the complete solution has been tested with users.]

So far, validation with users has been limited to the survey of 28 students described in Section 1.2, which informed the design of the storytelling component. A comparison study is planned in which one group of students uses a story lesson and another reads the textbook.

[Report who evaluated the complete solution, how many took part, how their feedback was collected, the main findings, the changes made as a result, and any concerns that remain.]

6.5 Achievement of Common Objectives

Table 6.7 states how far each group objective was achieved.

Table 6.7: Achievement of group objectives

Objective	Status	Justification

Objective	Status	Justification
G1. Shared curriculum knowledge base	Achieved	The textbooks were indexed, and three components retrieve from the shared index
G2. Four working components	Achieved	Each component is implemented and has component-level results
G3. One integrated application	Partly achieved	All four components run in one application and use shared services. End-to-end tests of the full flow are still to be completed.
G4. Openly available models through a common runtime	Partly achieved	Storytelling, Conversational AI and Knowledge Maps use open models through Ollama. The quiz database and the Knowledge Maps model depend on cloud services.
G5. Evaluation of components and the complete solution	Partly achieved	Each component was evaluated. Evaluation of the complete solution with users is still to be done.

6.6 Integrated Limitations and Threats

The complete solution has the following limitations and threats to validity.

- **No evaluation with users yet.** The results show that the components work technically. They do not yet show that students learn more.
- **Component results are not comparable.** Each component was evaluated in a different way, so the results cannot be combined into one figure for the solution.
- **Rated scores.** The model comparisons for the tutor and for Knowledge Maps use rated scores, which depend on who rates them and on the items chosen.
- **Limited data hand-off.** The components share content but pass little or no data to each other, so the flow relies on the student moving between them.
- **Dependence on shared services.** Three components depend on one model runtime. If it stops, all three stop.
- **Partly online.** The quiz database and the Knowledge Maps model need an internet connection, so the solution does not run fully offline.
- **One computer.** The solution has been run by one user at a time on one computer. Its behaviour with many students is not known.

- **One subject and one medium.** The content is Science in English, so the results may not hold for other subjects or for Sinhala and Tamil medium students.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

The solution offers a student one place to study a Science topic from first explanation to final revision, with content that follows the textbook. Table 7.1 lists the value for each group of beneficiaries.

Table 7.1: Value for each beneficiary

Beneficiary	Value
Students	Personal explanations, answers at any time, practice that adapts, and checked revision notes
Teachers and tuition teachers	A study aid that follows the syllabus, and a view of how students are doing in quizzes
Parents	Support at home that does not depend on paying for extra classes
Schools and tuition institutes	A tool that runs on their own computer, with low running cost
Society	A step towards more equal access to personal learning support

The evidence for this value is so far technical, together with the student survey. Section 6.4 sets out the validation with users that is still needed.

7.2 Market Alternatives and Differentiation

The target users are O/L students in Sri Lanka, reached directly or through schools and tuition institutes. Table 7.2 compares the solution with the alternatives a student has today.

Table 7.2: Alternatives and differentiation

Alternative	What it offers	Difference from Study Buddy
General AI chatbots	Answers to any question	Not tied to the Sri Lankan syllabus, and can give answers that are wrong

Alternative	What it offers	Difference from Study Buddy
Khanmigo [9]	A paid AI tutor tied to Khan Academy's own content	Its content is not the Sri Lankan syllabus, and it is a paid cloud service
e-thaksalawa [11]	The official online learning portal of the Ministry of Education	Provides lessons and learning material that are the same for every student. It has no AI tutoring and no personalised content
DP Education [12]	A free online learning platform	Offers video lessons with limited AI support. The content is not personalised to the student, and it does not cover the four study stages in one place
Private tuition	Teaching by a person, in a class	Costs a fee, and is not available at any time

The solution differs from these in three ways taken together: its content comes from the official textbooks, it covers four stages of studying in one application, and most of it runs on openly available models on a local computer.

7.3 Business Model and Feasibility

We propose a model in which the solution is free for individual students and is licensed to schools and tuition institutes, which would install it on their own computer.

The main costs are a computer able to run the models, installation, and maintenance of the content when a textbook changes. Because the main models run locally, there is no charge for each request.

The technical feasibility is supported by our results. A story lesson is generated in about 44 seconds on the development computer, and the expression model processes a frame in 27.7 milliseconds on a CPU. Possible partners are tuition institutes, schools and education bodies.

7.4 Adoption, Intellectual Property and Compliance

Adoption barriers. A school needs a computer able to run the models, and parts of the solution need an internet connection. Teachers must trust the content before they recommend it. Stories are written in English only, which excludes students who study in Sinhala or Tamil.

Ownership.

Licensing of third-party material. Three points need attention before any commercial use. The textbooks are official publications, so permission is needed to use their content in a product. The language models are released under their own licences, which set conditions on use. The YOLOv8 software is released under a licence that requires either that a product using it is made open source or that a commercial licence is bought.

Data protection. The users are school students, most of them under 18. The Personal Data Protection Act, No. 9 of 2022, applies to their personal data [10]. The storytelling survey collects no names. Adaptive Assessment uses the student's facial expression from a webcam and stores quiz data in a cloud database. This is more sensitive data, and it requires clear consent from students and parents, a stated purpose, and a limit on what is stored.

Ethics and safety. The content shown to students is restricted to the textbook as far as the methods allow, and each component has a fallback for failure. The solution is presented as a study aid and not as a source of official answers.

7.5 Sustainable Development Goal Contribution

The solution could contribute to Target 4.1 by helping secondary students to understand and retain Science content, and to Target 4.5 by offering personal support to students who cannot afford private tuition [8]. These contributions are plausible but unproven. Showing them would need a study of learning outcomes with a larger group of students, including students from less well-resourced schools.

The solution could also have negative effects. It needs a capable computer, so it may reach well-resourced schools first and widen the gap it is meant to narrow. Students

may rely on it in place of their teacher or their own effort. The use of a webcam raises a privacy risk for children.

Sustaining any benefit would need four measures: versions in Sinhala and Tamil, a version that runs on modest hardware, clear consent and data rules for the quiz, and a way to update the content whenever the textbooks change.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

The common problem was that O/L students lack learning support that is personal, accurate to the syllabus, and available through every stage of studying a topic. Our answer is that such support can be provided by one application, built from four components on a shared base of textbook knowledge.

The evidence supports this at the technical level. All four components were built and joined in one application. Each has a model that suits its task: the storytelling component produced a valid lesson in all 12 benchmark cases, the selected tutor model scored highest of three, the selected expression model runs in real time on a CPU, and the selected vision-language model scored highest of three. Three components share one textbook index and three share one model runtime.

The evidence does not yet show that students learn more with the solution. That question remains open until the complete solution is tested with students.

8.2 Principal Integrated Contribution

The principal contribution of the team, beyond the four components, is the integration. We showed that four different AI methods, covering story generation, tutoring by voice, adaptive assessment and the reading of handwritten notes, can be built on one base of curriculum knowledge and run in one application.

Two things follow from this. A student meets consistent content at every stage, because every component retrieves from the same textbooks. A team can also build such a solution with mostly openly available models on a local computer, which keeps the running cost low for a school.

8.3 Future Directions

The following work is needed, in order of priority.

- **Test the complete solution with students**, to find out whether it improves understanding and whether students complete the full flow.

- **Pass data between the components**, so that a story lesson leads to questions and a quiz on the same topic, and quiz results guide what the student studies next.
- **Complete the end-to-end and load tests**, including use by several students at once.
- **Resolve the licensing and data protection points** in Section 7.4 before any release.
- **Add Sinhala and Tamil**, and extend the content to other STEM subjects.
- **Reduce the hardware needed**, so that the solution runs in schools with modest computers.

REFERENCES

[1] D. Sun, G. Cheng, P. L. H. Yu, J. Jia, Z. Zheng, and A. Chen, "Personalized STEM education empowered by artificial intelligence: A comprehensive review and content analysis," Interactive Learning Environments, vol. 33, no. 7, pp. 4419–4441, 2025, doi: 10.1080/10494820.2025.2462156.

[2] C. Leon, J. Lipuma, and X. Oviedo-Torres, "Artificial intelligence in STEM education: A transdisciplinary framework for engagement and innovation," Frontiers in Education, vol. 10, art. no. 1619888, 2025, doi: 10.3389/feduc.2025.1619888.

[3] W. Villegas-Ch., D. Buenano-Fernandez, A. Maldonado Navarro, and A. Mera-Navarrete, "Adaptive intelligent tutoring systems for STEM education: Analysis of the learning impact and effectiveness of personalized feedback," Smart Learning Environments, vol. 12, art. no. 41, 2025, doi: 10.1186/s40561-025-00389-y.

[4] Hariyanto, F. X. D. Kristianingsih, and R. Maharani, "Artificial intelligence in adaptive education: A systematic review of techniques for personalized learning," Discover Education, vol. 4, art. no. 458, 2025, doi: 10.1007/s44217-025-00908-6.

[5] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," arXiv:2005.11401, 2020.

[6] N. N. Varghese, B. Jose, T. Bindhumol, A. Cleetus, and S. B. Nair, "The power duo: Unleashing cognitive potential through human-AI synergy in STEM and non-STEM education," Frontiers in Education, vol. 10, art. no. 1534582, 2025, doi: 10.3389/feduc.2025.1534582.

[7] F. Gong, "Design and implementation of an intelligent educational interaction system with integrated multimodal emotion recognition and adaptive content delivery," Discover Artificial Intelligence, vol. 6, art. no. 48, 2026, doi: 10.1007/s44163-025-00671-5.

[8] United Nations, "Transforming our world: The 2030 Agenda for Sustainable Development," General Assembly resolution A/RES/70/1, 2015.

[9] Khan Academy, "Khanmigo for learners." Accessed: Oct. 11, 2026. [Online]. Available: https://khanmigo.ai/learners

[10] Parliament of the Democratic Socialist Republic of Sri Lanka, Personal Data Protection Act, No. 9 of 2022.

[11] Ministry of Education, Sri Lanka, "e-thaksalawa.".

[12] DP Education, "DP Education." .

APPENDICES

The appendices hold the material needed to verify the integrated solution.

Appendix A Integrated Interface Specification

Table A.1 specifies the requests of the storytelling component. The requests of the other three components are to be added in the same form.

Table A.1: Requests of the storytelling component (address prefix /narrative-learning)

Request	Receives	Returns
Status	Nothing	Whether the component is ready, and whether the trained classifier is loaded
Survey	Nothing	The six survey questions and their options
Classify	The student's survey answers	The story theme
Chapters	Nothing	The textbooks and their chapters
Generate	The topic, the theme, the interest and the level of struggle	The lesson: explanation, story, key definitions, equations, exam points and source passages
Theme feedback	Whether the theme matched the student	A confirmation

- PENDING [Add one table for each of the other three components: Conversational AI, Adaptive Assessment and Knowledge Maps.]

Appendix B Team Contribution Declaration

Table B.1 records the contribution of each member as agreed by the team.

Table B.1: Team contribution declaration

Member	Individual contribution	Shared contribution	Signature
Elwitigala S.A.I.	Designed, built and evaluated RAG-Based Personalized STEM Storytelling	Joined the component to the common application; linked it to the shared textbook index; compiled the common report	
Manodya K.M.B.	Designed, built and evaluated Multilingual Conversational AI	Joined the component to the common application; prepared the shared textbook index	
Basnayaka W.B.M.D. K.	Designed, built and evaluated Emotion-Aware Adaptive Assessment	Joined the component to the common application; set up the quiz database, sign-in and question generation service	
Dilshan W.A.	Designed, built and evaluated Handwritten Content Analysis and Synthesis	Joined the component to the common application; set up the vision-language model and the optional fallback model	